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### **THERMAL MANAGEMENT COMPONENT, THERMAL MANAGEMENT ASSEMBLY, BATTERY, AND ELECTRIC DEVICE**

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#### **Abstract**

Provided are a thermal management component, a thermal management assembly, a battery, and an electric device. The thermal management component includes two straight walls and two bent walls. The two straight walls are arranged oppositely along a first direction. Each of the two bent walls is connected to one end of the straight walls along a second direction, where the second direction is perpendicular to the first direction. The two bent walls and the two straight walls enclose to form a holding space, and the holding space is configured to contain a heat exchange medium.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Patent Application No. PCT/CN2023/073607, filed on Jan. 28, 2023, which is incorporated herein by reference in its entirety.

### TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of batteries, and particularly to a thermal management component, a thermal management assembly, a battery, and an electric device.

### BACKGROUND ART

[0003] In recent years, new energy vehicles have experienced a leapfrog development. In the field of electric vehicles, power batteries serve as the energy source for electric vehicles and play an irreplaceable important role. The battery consists of a box and multiple battery cells accommodated within the box. The battery, as the core component of new energy vehicles, has high requirements in terms of both safety and service life. The battery cells within the battery generate a large amount of heat during continuous charging and discharging processes. Thus, a thermal management component, configured for adjusting the temperature of the battery cells, is arranged in the box. However, the existing thermal management components have manufacturing tolerances and relatively poor production quality, leading to poor assembly precision between the thermal management component and the battery cells, which is unfavorable for adjusting the temperature of the battery cells.

### SUMMARY

[0004] The embodiments of the present disclosure provide a thermal management component, a thermal management assembly, a battery, and an electric device, which is capable of effectively improving the manufacturing precision of the thermal management component.

[0005] In a first aspect, the embodiments of the present disclosure provide a thermal management component, including two straight walls and two bent walls. The two straight walls are arranged oppositely along a first direction. Each of the two bent walls is connected to one end of the straight walls along a second direction, where the second direction is perpendicular to the first direction. The two bent walls and the two straight walls enclose to form a holding space, and the holding space is configured to contain a heat exchange medium.

[0006] In the above technical solution, the thermal management component is provided with two straight walls arranged oppositely along the first direction, and the same ends of the two straight walls are connected by a bent wall, so as to form a holding space configured for containing the heat exchange medium. This enables the thermal management component to perform temperature management. The thermal management component with this structure facilitates the deformation of the bent wall when compressing the thermal management component along the first direction during manufacturing, thereby allowing absorption of the manufacturing tolerances of the thermal management component through compression. Therefore, the size of the thermal management component in the first direction is controlled, effectively improving the manufacturing precision of the thermal management component and ensuring the assembly precision between the thermal management component and the battery cells, thus favoring the temperature adjustment of the battery cells by the thermal management component.

[0007] In some embodiments, along the second direction, the bent wall bends in a direction away from the holding space.

[0008] In the above technical solution, by arranging the bent wall to bend in a direction away from the holding space along the second direction, meaning that the bent wall protrudes outward at both

ends of the thermal management component, this structure, on the one hand, facilitates manufacturing, helps reduce the difficulty of manufacturing the thermal management component, and on the other hand, effectively alleviates the phenomenon of rupture at the connection between the bent wall and the straight wall when compressing the thermal management component to absorb manufacturing tolerances.

[0009] In some embodiments, the bent wall includes a first segment and two second segments. A second segment, a first segment, and the other second segment are sequentially connected, and the two second segments are respectively connected to the two straight walls. The second segment is arranged obliquely relative to the straight wall connected to it, and along the first direction, the first segment is positioned between the two straight walls.

[0010] In the above technical solution, the bent wall is provided with second segments that are respectively connected to the two straight walls and a first segment connected between the two second segments. The second segments bend in the first direction toward the other second segment, such that the distance between the two second segments gradually decreases in the second direction toward the first segment, thus allowing the bent wall to form a contracted structure. Thus, when compressing the thermal management component to absorb manufacturing tolerances, the second segments can deform relative to the first segment, to facilitate the compression of the thermal management component to absorb manufacturing tolerances.

[0011] In some embodiments, a wall thickness of the straight wall is  $L_{sub.1}$ , and a wall thickness of the first segment is  $L_{sub.2}$ , satisfying a condition:  $L_{sub.2} > L_{sub.1}$ .

[0012] In the above technical solution, by setting the wall thickness of the first segment of the bent wall to be greater than the wall thickness of the straight wall, the structural strength and the deformation resistance of the first segment of the bent wall are increased. This effectively alleviates the occurrence of deformation or excessive curvature change of the first segment of the bent wall when compressing the thermal management component to absorb manufacturing tolerances, which is conducive to maintaining the original structural shape of the first segment of the bent wall. Consequently, it can reduce the occurrence of excessive compression or crushing of the thermal management component, thereby lowering the risk of the bent wall being damaged or cracked during the compression process of the thermal management component.

[0013] In some embodiments, along the second direction, the first segment is an arc-shaped structure bending in a direction away from the holding space.

[0014] In the above technical solution, by arranging the first segment as an outwardly bending arc-shaped structure, on the one hand, it facilitates the connection between the first segment and the second segment, thus reducing the difficulty of manufacturing; on the other hand, it enables a smooth arc transition between the first segment and the second segment, which alleviates the phenomenon of excessively small curvature radius at the connection between the first segment and the second segment. This effectively reduces the risk of the connection between the first segment and the second segment being damaged or cracked during the compression process of the thermal management component.

[0015] In some embodiments, the wall thickness of the second segment gradually increases from the end connected to the straight wall toward the end connected to the first segment.

[0016] In the above technical solution, by setting the wall thickness of the second segment to gradually increase from the end connected to the straight wall to the end connected to the first segment, it facilitates manufacturing the first segment as a structure in which a wall thickness is greater than the wall thickness of the straight wall, thus reducing manufacturing difficulty.

[0017] In some embodiments, the thermal management component further includes a reinforcement portion, and the reinforcement portion is arranged between the two second segments along the first direction.

[0018] In the above technical solution, by arranging a reinforcement portion between the two second segments along the first direction, the reinforcement portion can provide certain support for

the two second segments, thereby increasing the deformation resistance of the first segment of the bent wall. This effectively alleviates the occurrence of deformation or excessive curvature change of the first segment of the bent wall when compressing the thermal management component to absorb manufacturing tolerances, which is conducive to maintaining the original structural shape of the first segment of the bent wall. Consequently, it can reduce the occurrence of excessive compression or crushing of the thermal management component, thereby lowering the risk of the bent wall being damaged or cracked during the compression process of the thermal management component.

[0019] In some embodiments, along the first direction, one end of the reinforcement portion is connected to one of the two second segments, and the other end is arranged at intervals with the other of the two second segments.

[0020] In the above technical solution, by arranging the reinforcement portion to be connected to one second segment and spaced from the other second segment, the reinforcement portion can provide support to the two second segments and reserve compression allowance at the same time, thereby facilitating the compression of the thermal management component to absorb manufacturing tolerances.

[0021] In some embodiments, along the first direction, each end of the reinforcement portion is connected to one of the two second segments.

[0022] In the above technical solution, by connecting each end of the reinforcement portion with each of the two second segments, this structure effectively enhances the connection strength between the reinforcement portion and the bent wall, which is beneficial for improving the supporting effect of the reinforcement portion on the two second segments.

[0023] In some embodiments, along the second direction, the reinforcement portion is arranged at intervals with the first segment.

[0024] In the above technical solution, by arranging the reinforcement portion at intervals with the first segment along the second direction, a gap is provided between the reinforcement portion and the first segment in the second direction, which helps to reduce the manufacturing difficulty of the thermal management component.

[0025] In some embodiments, the thermal management component further includes multiple partition portions. The multiple partition portions are arranged at intervals along the second direction. The partition portion is arranged obliquely between the two straight walls and connected to the two straight walls. The multiple partition portions are configured to divide the holding space into multiple medium flow channels, and the medium flow channels are configured to contain a heat exchange medium.

[0026] In the above technical solution, by arranging multiple partition portions at intervals along the second direction within the holding space of the thermal management component, the holding space is divided into multiple medium flow channels configured for the flow of heat exchange medium. This facilitates the control of the flow rate and flow volume of the heat exchange medium within the thermal management component. Additionally, by arranging the partition portion in an inclined manner and connecting it between the two straight walls, the partition portion can deform more easily when the thermal management component is compressed along the first direction to absorb manufacturing tolerances, which helps to reduce the difficulty of compressing the thermal management component.

[0027] In some embodiments, the partition portion is provided with an opposing first surface and second surface along its thickness direction. The two straight walls include a first straight wall and a second straight wall, wherein the first straight wall is provided with a third surface facing the medium flow channel, and the second straight wall is provided with a fourth surface facing the medium flow channel; the first surface is arranged at an acute angle with the third surface and at an obtuse angle with the fourth surface, and the second surface is arranged at an obtuse angle with the third surface and at an acute angle with the fourth surface, wherein the first surface is connected to

the third surface through a first chamfered surface, and/or the second surface is connected to the fourth surface through a second chamfered surface.

[0028] In the above technical solution, the two surfaces of the partition portion that form acute angles with the straight wall are connected through chamfered surfaces, which increases the thickness of the portion where the partition portion connects with the straight wall along the second direction. On the one hand, it enhances the stability and reliability of the connection between the partition portion and the straight wall, and on the other hand, it also helps to alleviate the occurrence of cracks between the partition portion and the straight wall.

[0029] In some embodiments, along the first direction, a support portion is provided in a protruding manner on a side of the straight wall facing the holding space, and a size of the support portion is smaller than a distance between the two straight walls.

[0030] In the above technical solution, by providing a support portion on the side of the straight wall facing the holding space, and with the size of the support portion along the first direction being smaller than the distance between the two straight walls, the support portion is arranged on one straight wall and is arranged at intervals with the other straight wall in the first direction. The thermal management component with this structure allows the support portion to provide some support to the two straight walls when the thermal management component is compressed to absorb manufacturing tolerances, thereby alleviating the phenomenon of excessive compression or crushing of the thermal management component. This is beneficial for maintaining the smooth flow of the heat exchange medium within the holding space.

[0031] In some embodiments, both straight walls are provided with support portions, and along the first direction, the projections of the support portions on the two straight walls do not overlap.

[0032] In the above technical solution, by providing support portions on both straight walls, where the projections of the support portions on the two straight walls along the first direction do not overlap, the supporting effect of the support portions on the two straight walls is enhanced. This further reduces the risk of excessive compression or crushing of the thermal management component.

[0033] In a second aspect, the embodiments of the present disclosure further provide a thermal management assembly, including a first current collector, a second current collector, and the above-mentioned thermal management component. Along a third direction, the first current collector and the second current collector are respectively connected to both ends of the thermal management component. The first direction, the second direction, and the third direction are perpendicular to each other in pairs. The first current collector and the second current collector are respectively provided with a medium inlet and a medium outlet, wherein the medium inlet and the medium outlet are in communication with the holding space.

[0034] In the above technical solution, the thermal management assembly is further provided with a first current collector and a second current collector, which are connected to both ends of the thermal management component along the third direction. The first current collector and the second current collector are respectively provided with a medium inlet and a medium outlet. Therefore, by communicating the two ends of the holding space respectively with the medium inlet and medium outlet, the heat exchange medium can flow into or out of the holding space, thereby enabling the thermal management function of the thermal management assembly.

[0035] In a third aspect, the embodiments of the present disclosure further provide a battery, including a box, a battery cell, and the above-mentioned thermal management assembly. The thermal management assembly is arranged within the box, and the thermal management assembly is configured to regulate the temperature of the battery cell.

[0036] In some embodiments, the battery cell is provided with a first side surface, wherein the first side surface is a surface with the largest area among the outer surfaces of the battery cell, and the first side surface abuts the straight wall.

[0037] In the above technical solution, by arranging the straight wall of the thermal management

component to abut the first side surface of the battery cell, that is, by arranging the thermal management component on the side of the battery cell with the largest surface area, sufficient heat exchange area between the battery cell and the thermal management component can be ensured. This is beneficial for enhancing the thermal management capability of the thermal management component for the battery cell, thereby effectively reducing the safety risks caused by temperature rise during the use of the battery.

[0038] In the fourth aspect, the embodiments of the present disclosure further provide an electric device comprising the battery as described above, wherein the battery is configured to provide electrical energy.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0039] To more clearly illustrate the technical solutions of the embodiments of the present disclosure, the following will briefly introduce the drawings used in the embodiments. It should be understood that the following drawings only show some embodiments of the present disclosure, and therefore they should not be regarded as a limitation on the scope. Those ordinary skilled in the art can also obtain other related drawings based on these drawings without inventive effort.

[0040] FIG. 1 is a schematic diagram of a structure of a vehicle provided in some embodiments of the present disclosure;

[0041] FIG. 2 is an exploded view of the structure of a battery provided in some embodiments of the present disclosure;

[0042] FIG. 3 is a schematic diagram of an assembly of a thermal management assembly and a battery cell provided in some embodiments of the present disclosure;

[0043] FIG. 4 is a schematic diagram of a structure of a thermal management assembly provided in some embodiments of the present disclosure;

[0044] FIG. 5 is a schematic diagram of a structure of a thermal management component provided in some embodiments of the present disclosure;

[0045] FIG. 6 is a partially enlarged view at A of a thermal management component shown in FIG. 5;

[0046] FIG. 7 is a front view of a thermal management component in a third direction provided in some embodiments of the present disclosure;

[0047] FIG. 8 is a partially enlarged view at B of a thermal management component shown in FIG. 7;

[0048] FIG. 9 is a front view of a thermal management component in a third direction provided in some other embodiments of the present disclosure;

[0049] FIG. 10 is a partially enlarged view at C of a thermal management component shown in FIG. 9;

[0050] FIG. 11 is a front view of a thermal management component in a third direction provided in some other embodiments of the present disclosure;

[0051] FIG. 12 is a partially enlarged view at D of a thermal management component shown in FIG. 11;

[0052] FIG. 13 is a partially enlarged view at E of a thermal management component shown in FIG. 7; and

[0053] FIG. 14 is a front view of a thermal management component in a third direction provided in yet some other embodiments of the present disclosure.

[0054] Reference numerals: **1000**—vehicle; **100**—battery; **10**—box; **11**—first box body; **12**—second box body; **20**—battery cell; **21**—first side surface; **30**—thermal management assembly; **31**—thermal management component; **311**—straight wall; **3111**—first straight wall; **3111a**—third

surface; **3112**—second straight wall; **3112a**—fourth surface; **312**—bent wall; **3121**—first segment; **3122**—second segment; **313**—holding space; **3131**—medium flow channel; **314**—reinforcement portion; **315**—partition portion; **3151**—first surface; **3152**—second surface; **316**—first chamfered surface; **317**—second chamfered surface; **318**—support portion; **32**—first current collector; **321**—medium inlet; **33**—second current collector; **331**—medium outlet; **200**—controller; **300**—motor; X—first direction; Y—second direction; Z—third direction.

#### DETAILED DESCRIPTION OF EMBODIMENTS

[0055] In order to make the objective, technical solution, and advantages of the embodiments of the present disclosure clearer, the following will provide a clear and complete description of the technical solution in the embodiments of the present disclosure, in conjunction with the drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only a part of the embodiments of the present disclosure, rather than all embodiments. Based on the embodiments in the present disclosure, all other embodiments obtained by those of ordinary skill in the art without making inventive efforts are within the scope of protection of the present disclosure.

[0056] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as understood by those skilled in the art of the relevant technical field. The terms used in the summary of the present disclosure are used for the purpose of describing specific embodiments and are not intended to limit the present disclosure. The terms, “comprise”, “provide” and their variations, and any other variations, are intended to cover non-exclusive inclusion in the summary, the claims, and the above drawings of the present disclosure. The terms “first”, “second”, and so on in the summary and claims of the present disclosure or in the above drawings are used to distinguish between different objects and are not intended to describe a specific sequence or hierarchy.

[0057] The term “embodiment” used in the present disclosure means that specific features, structures, or characteristics described in conjunction with the embodiments can be included in at least one embodiment of the present disclosure. The use of the phrase in various parts of the summary does not necessarily refer to the same embodiment, nor does it imply that these embodiments are mutually exclusive or independent of each other.

[0058] In the description of the present disclosure, it should be noted that, unless otherwise clearly stipulated and limited, the terms “provide”, “interconnect”, “connect”, and “attach” should be understood in a broad sense. For example, it can be a fixed connection, a detachable connection, or an integral connection; and it can be a direct connection, an indirect connection through an intermediary, or an internal communication between two components. Those of ordinary skill in the art can understand the meanings of the above terms in the present disclosure according to specific situations.

[0059] In the present disclosure, the term “and/or” is used as a way to describe the relationship between associated objects, indicating that there can be three possible relationships. For example, “A and/or B” can represent: the presence of A, the presence of both A and B, or the presence of B. Additionally, the character “/” in the present disclosure generally represents an “or” relationship between the preceding and following associated objects.

[0060] In the embodiments of the present disclosure, identical reference numerals indicate the same components. For the sake of conciseness, detailed explanations of the same components are omitted in different embodiments. It should be understood that the dimensions such as thickness, length, and width of various components in the embodiments of the present disclosure as shown in the drawings, and the overall dimensions such as thickness, length, and width of the integrated device, are provided as illustrative examples and should not be construed to limit the scope of the present disclosure.

[0061] In the present disclosure, the term “multiple” refers to two or more (including two).

[0062] In the present disclosure, the battery cell can comprise various types of batteries such as lithium-ion secondary batteries, lithium-ion primary batteries, lithium-sulfur batteries, sodium-

lithium ion batteries, sodium-ion batteries, and magnesium-ion batteries. The embodiments of the present disclosure are not limited to any specific type of battery. The battery cell can have various shapes such as cylindrical, flat, rectangular, or other shapes. The embodiments of the present disclosure are not limited in respect thereof either. Battery cells are generally categorized into three types according to the encapsulation method: cylindrical battery cells, prismatic battery cells, and soft pouch battery cells. The embodiments of the present disclosure are not limited herein either. [0063] The batteries mentioned in the embodiments of the present disclosure refer to a single physical module that comprises one or multiple battery cells to provide higher voltage and capacity. For example, the batteries mentioned in the present disclosure can comprise battery modules or battery packs, and so on. A battery generally includes a box for encapsulating one or multiple battery cells or multiple battery modules. The box helps to prevent liquids or other foreign substances from affecting the charging or discharging of the battery cells.

[0064] The battery cell comprises a housing, an electrode assembly, and an electrolyte, and the housing is configured to accommodate the electrode assembly and the electrolyte. The electrode assembly is composed of a positive electrode sheet, a negative electrode sheet, and a separator. The battery cells work primarily on the movement of metal ions between the positive electrode sheet and the negative electrode sheet.

[0065] In recent years, new energy vehicles have experienced a leapfrog development. In the field of electric vehicles, power batteries serve as the energy source for electric vehicles and play an irreplaceable important role. The battery consists of a box and multiple battery cells accommodated within the box. The battery, as the core component of new energy vehicles, has high requirements in terms of both safety and cycle service life.

[0066] The inventors have found that, in conventional power batteries, to ensure that the battery obtains sufficient power, multiple battery cells within the box of the battery are typically arranged in a stacked manner. However, during the continuous charging and discharging process, the battery cells inside the battery generate a large amount of heat, leading to a rapid increase in the internal temperature of the battery. The stacked arrangement of multiple battery cells exacerbates this phenomenon. This significantly affects the performance and lifespan of the battery, and may even pose significant safety hazards during use, which is not conducive to consumer safety. Therefore, a thermal management component is usually arranged inside the battery, the thermal management component is arranged on one side of the battery cell, and a channel for the flow of heat exchange medium is formed inside the thermal management component, thereby enabling the cooling or heating of the battery cell. However, the thermal management component with the structure has a large manufacturing tolerance, which is not conducive to ensuring the assembly precision between the thermal management component and the battery cell.

[0067] Based on the above considerations, in order to solve the problem of the low manufacturing precision of existing thermal management components, the inventors, after in-depth research, designed a thermal management component, wherein the thermal management component includes two straight walls and two bent walls, the two straight walls are arranged oppositely along a first direction. Each of the two bent walls is connected to one end of the straight walls, along a second direction perpendicular to the first direction. The two bent walls and the two straight walls enclose to form a holding space, and the holding space is configured to contain a heat exchange medium.

[0068] In the thermal management component with this structure, the thermal management component is provided with two straight walls arranged oppositely along the first direction, and the same ends of the two straight walls are connected by a bent wall, so as to form a holding space configured for containing the heat exchange medium. This enables the thermal management component to perform temperature management. The thermal management component with this structure facilitates the deformation of the bent wall when compressing the thermal management component along the first direction during manufacturing, thereby allowing absorption of the manufacturing tolerances of the thermal management component through compression. Therefore,



the size of the thermal management component in the first direction is controlled, effectively improving the manufacturing precision of the thermal management component and ensuring the assembly precision between the thermal management component and the battery cells, thus favoring the temperature adjustment of the battery cells by the thermal management component. [0069] The thermal management component disclosed in the embodiments of the present disclosure can be used in various electric devices, including but not limited to vehicles, ships, or aircraft. The power system of the electric device can be composed using the thermal management system, battery, and other components disclosed in the present disclosure. This is beneficial for reducing the manufacturing tolerance of the thermal management component, thereby improving the assembly precision between the thermal management component and the battery cell.

[0070] The embodiments of the present disclosure provide an electric device that uses batteries as a power source. The electric device can comprise, but is not limited to, mobile phones, tablets, laptops, electric toys, power tools, electric bicycles, electric cars, boats, spacecraft, and more. The electric toys can include stationary or mobile electric toys, for example, video game consoles, electric car toys, electric boat toys, and electric airplane toys. The spacecraft can include aircraft, rockets, space shuttles, and spacecraft, among others.

[0071] For ease of explanation, the following embodiment will be described using one embodiment of the present disclosure where the electric device is a vehicle **1000** as an example.

[0072] Referring to FIG. 1, FIG. 1 is a schematic diagram of a structure of a vehicle **1000** provided in some embodiments of the present disclosure. The vehicle **1000** can be a gasoline-powered car, a natural gas car, or a new energy vehicle. The new energy vehicles can be pure electric cars, hybrid cars, extended-range cars, and so on. A battery **100** is provided inside the vehicle **1000**, wherein the battery **100** can be arranged at the bottom, front, or rear of the vehicle **1000**. The battery **100** is configured to provide power for the vehicle **1000**. For instance, the battery **100** can be the operational power source for the vehicle **1000**. Additionally, the vehicle **1000** also comprises a controller **200** and a motor **300**. The controller **200** is configured to manage the supply of power from the battery **100** to the motor **300**, for example, for working power requirements for starting, navigating, and driving the vehicle **1000**.

[0073] In some embodiments of the present disclosure, the battery **100** serves not only as the operational power source for the vehicle **1000** but also as the driving power source for the vehicle **1000**. It can replace or partially replace gasoline or natural gas to provide driving propulsion for the vehicle **1000**.

[0074] Referring to FIG. 2, FIG. 2 is an exploded view of the structure of a battery **100** provided in some embodiments of the present disclosure. The battery **100** comprises a box **10** and multiple battery cells **20**, wherein the battery cells **20** are configured to be housed in the box **10**.

[0075] The box **10** is configured to provide an assembly space for the battery cell **20**, and the box **10** can be of various structures. In some embodiments, the box **10** can include a first box body **11** and a second box body **12**, wherein the first box body **11** and the second box body **12** are capped together. The first box body **11** and the second box body **12** collectively define an assembly space configured for accommodating the battery cell **20**. The second box body **12** can be a hollow structure open at one end, and the first box body **11** can be a plate-like structure. The first box body **11** is capped over an open side of the second box body **12** so that the first box body **11** and the second box body **12** together define an assembly space. Or, both the first box body **11** and the second box body **12** can be hollow structures open at one end, and an open side of the first box body **11** is capped over an open side of the second box body **12**. Of course, the box **10** formed by the first box body **11** and the second box body **12** can be of a variety of shapes, such as cylindrical or rectangular.

[0076] In the battery module **100**, the multiple battery cells **20** can be connected in series, parallel, or hybrid. A hybrid connection refers to having both series and parallel connections among multiple battery cells **20**. The multiple battery cells **20** can be directly connected in series, in parallel, or in a

hybrid connection. Then, the whole comprising the multiple battery cells **20** is housed in the box **10**. Of course, the battery **100** can also be in the form that, the multiple battery cells **20** are first connected in series, in parallel, or in hybrid connection to form battery modules, and then the multiple battery modules are connected in series, in parallel, or in hybrid connection to be formed into a whole, which is then accommodated within the box **10**. The battery **100** can also comprise other structures. For example, the battery **100** can also comprise a busbar component, which is configured for achieving electrical connections between the multiple battery cells **20**.

[0077] In one embodiment, the battery **100** can also include a thermal management system, and the thermal management system is arranged within the box **10**. The thermal management system can include multiple thermal management assemblies **30**, and the thermal management assemblies **30** are configured to manage the temperature of the battery cell **20**, thereby cooling down or heating up the battery cell **20**.

[0078] With reference to FIG. 2, and with further reference to FIG. 3, FIG. 3 is a schematic diagram of an assembly of a thermal management assembly **30** and a battery cell **20** provided in some embodiments of the present disclosure. The battery **100** includes multiple rows of battery cells **20**, with the multiple rows of battery cells **20** arranged along the first direction X. Each row of battery cells **20** includes multiple battery cells **20** arranged along the third direction Z. It should be noted that in other embodiments, the battery **100** can also include only multiple battery cells **20** arranged along the first direction X, meaning that the battery **100** includes only a single row of multiple battery cells **20** arranged along the first direction X.

[0079] Each battery cell **20** can be a secondary battery or a primary battery; and it can also be a lithium-sulfur battery, sodium-ion battery, magnesium-ion battery, or other types, but is not limited herein. The battery cell **20** can have various shapes such as cylindrical, flat, rectangular, or other shapes. For example, in FIG. 3, the shape of the battery cell **20** is a rectangular prism. Accordingly, the first direction X is the thickness direction of the battery cell **20**, and the third direction Z is the length direction of the battery cell.

[0080] In some embodiments, the thermal management system includes multiple thermal management assemblies **30**. Along the first direction X, a thermal management assembly **30** is arranged between two adjacent rows of battery cells **20**. The thermal management assembly **30** is configured to perform heat exchange with the battery cell **20** to manage the temperature of the battery cell **20**.

[0081] With reference to FIG. 3, and with further reference to FIG. 4, FIG. 4 is a schematic diagram of a structure of a thermal management assembly **30** provided in some embodiments of the present disclosure. The thermal management assembly **30** includes a thermal management component **31**, a first current collector **32**, and a second current collector **33**. Along the third direction Z, the first current collector **32** and the second current collector **33** are respectively connected to both ends of the thermal management component **31**.

[0082] The first direction X is the thickness direction of the thermal management component **31**, the second direction Y is the width direction of the thermal management component **31**, and the third direction Z is the length direction of the thermal management component **31**, which is also the extending direction of the thermal management component **31**.

[0083] The first current collector **32** is provided with a medium inlet **321**, and the medium inlet **321** is configured for the inflow of the heat exchange medium. The second current collector **33** is provided with a medium outlet **331**, and the medium outlet **331** is configured for the outflow of the heat exchange medium. Both the medium inlet **321** and the medium outlet **331** are in communication with the interior of the thermal management component **31** to facilitate the flow of the heat exchange medium within the thermal management component **31**, thereby enabling heat exchange with the battery cell **20** to manage the temperature of the battery cell **20**.

[0084] In one embodiment, the multiple thermal management assemblies **30** in the thermal management system can be structured in series with each other or in parallel with each other. If the

medium inlets **321** of the first current collectors **32** of multiple thermal management assemblies **30** are communicated with each other, and the medium outlets **331** of the second current collectors **33** of the multiple thermal management assemblies **30** are communicated with each other, then the multiple thermal management assemblies **30** are structured in parallel with each other. If the medium inlet **321** of the first current collector **32** of a thermal management assembly **30** of the multiple thermal management assemblies **30** is sequentially communicated with the medium outlet **331** of the second current collector **33** of another thermal management assembly **30**, then the multiple thermal management assemblies **30** are structured in series with each other.

[0085] It should be noted that in the structure where multiple thermal management assemblies **30** are interconnected in series or parallel, one thermal management assembly **30** can be directly connected to another thermal management assembly **30**; or, they can be connected through other components, such as connecting pipes, to enable a series structure or a parallel structure of multiple thermal management assemblies **30**.

[0086] Exemplarily, the heat exchange medium can be a gas, such as air or hydrogen; or, the heat exchange medium can be a liquid, such as water, saline solution, or liquid nitrogen. Of course, in other embodiments, the heat exchange medium can also be a solid, such as paraffin, which can achieve the heat exchange function through a phase change of the heat exchange medium. For example, when paraffin transitions from a solid state to a liquid state, it can absorb heat, thereby achieving a cooling effect on the battery cell **20**.

[0087] According to some embodiments of the present disclosure, with reference to FIGS. **3** and **4**, and with further reference to FIGS. **5**, **6**, and **7**, FIG. **5** is a schematic diagram of a structure of a thermal management component **31** provided in some embodiments of the present disclosure; FIG. **6** is a partially enlarged view at A of a thermal management component **31** shown in FIG. **5**; and FIG. **7** is a front view of a thermal management component **31** in a third direction Z provided in some embodiments of the present disclosure. The present disclosure provides a thermal management component **31**, wherein the thermal management component **31** includes two straight walls **311** and two bent walls **312**. The two straight walls **311** are arranged oppositely along a first direction X. Each of the two bent walls **312** is connected to one end of the straight walls **311** along a second direction Y, where the second direction Y is perpendicular to the first direction X. The two bent walls **312** and the two straight walls **311** enclose to form a holding space **313**, and the holding space **313** is configured for containing a heat exchange medium.

[0088] The two straight walls **311** are arranged opposite each other along the first direction X, meaning that the two straight walls **311** are arranged at intervals along the first direction X and oriented towards each other. The first direction X is the thickness direction of the straight walls **311**, meaning that the two straight walls **311** are arranged at intervals and parallel to each other.

[0089] Each of the two bent walls **312** is connected to one end of the straight walls **311** along a second direction Y, and the two bent walls **312** and the two straight walls **311** enclose to form a holding space **313**, which means that two straight walls **311** are connected at the same end by a bent wall **312**. In other words, one straight wall **311**, one bent wall **312**, the other straight wall **311**, and the other bent wall **312** are sequentially connected in a manner of head to tail to form a holding space **313** configured for containing a heat exchange medium. This allows the heat exchange medium to flow within the holding space **313**, thereby regulating the temperature of the battery cell **20**.

[0090] The bent wall **312** is a structure connected between two straight walls **311** and curved, and the bending direction of the bent wall **312** can vary. For instance, the bent wall **312** can curve in the second direction Y away from the other bent wall **312**, or it can curve in the second direction Y towards the other bent wall **312**.

[0091] The thermal management component **31** is provided with two straight walls **311** arranged oppositely along the first direction X, and the same ends of the two straight walls **311** are connected by a bent wall **312**, so as to form a holding space **313** configured for containing the heat exchange

medium. This enables the thermal management component **31** to perform temperature management. The thermal management component **31** with this structure facilitates the deformation of the bent wall **312** when compressing the thermal management component **31** along the first direction X during manufacturing, thereby allowing absorption of the manufacturing tolerances of the thermal management component **31** through compression. Therefore, the size of the thermal management component **31** in the first direction X is controlled, effectively improving the manufacturing precision of the thermal management component **31** and ensuring the assembly precision between the thermal management component **31** and the battery cells **20**, thus favoring the temperature adjustment of the battery cells **20** by the thermal management component **31**.

[0092] In some embodiments, referring to FIGS. **6** and **7**, along the second direction Y, the bent walls **312** bend in a direction away from the holding space **313**.

[0093] In other words, one bent wall **312** curves in the second direction Y away from the other bent wall **312**, causing the bent wall **312** to protrude outward at both ends of the thermal management component **31**.

[0094] By arranging the bent walls **312** to bend in a direction away from the holding space **313** along the second direction Y, this structure, on the one hand, facilitates manufacturing, helps reduce the difficulty of manufacturing the thermal management component **31**, and on the other hand, effectively alleviates the phenomenon of rupture at the connection between the bent wall **312** and the straight wall **311** when compressing the thermal management component **31** to absorb manufacturing tolerances.

[0095] According to some embodiments of the present disclosure, with reference to FIGS. **6** and **7**, and with further reference to FIG. **8**, FIG. **8** is a partially enlarged view at B of a thermal management component **31** shown in FIG. **7**. The bent wall **312** includes a first segment **3121** and two second segments **3122**. A second segment **3122**, a first segment **3121**, and the other second segment **3122** are sequentially connected, and the two second segments **3122** are respectively connected to the two straight walls **311**. The second segment **3122** is arranged obliquely relative to the straight wall **311** connected to it, and along the first direction X, the first segment **3121** is positioned between the two straight walls **311**.

[0096] A second segment **3122**, a first segment **3121**, and another second segment **3122** are connected in sequence, with the two second segments **3122** connected to the two straight walls **311**, respectively. In other words, the first segment **3121** is connected at each end in the first direction X to a straight wall **311** through a second segment **3122**.

[0097] The second segment **3122** is arranged obliquely relative to the straight wall **311** connected to it, and along the first direction X, the first segment **3121** is positioned between the two straight walls **311**. In other words, the second segment **3122** is an inclined structure arranged at an obtuse angle with respect to the straight wall **311**, and the second segment **3122** inclines in the first direction X toward the other second segment **3122**, thus allowing the first segment **3121** to be located between the two straight walls **311** in the first direction X.

[0098] The bent wall **312** is provided with second segments **3122** that are respectively connected to the two straight walls **311** and a first segment **3121** connected between the two second segments **3122**. The second segments **3122** bend in the first direction X toward the other second segment **3122**, such that the distance between the two second segments **3122** gradually decreases in the second direction Y toward the first segment **3121**, thus allowing the bent wall **312** to form a contracted structure. Thus, when compressing the thermal management component **31** to absorb manufacturing tolerances, the second segment **3122** can deform relative to the first segment **3121**, to facilitate the compression of the thermal management component **31** to absorb manufacturing tolerances.

[0099] According to some embodiments of the present disclosure, with reference to FIG. **7** and FIG. **8**, a wall thickness of the straight wall **311** is  $L_{sub.1}$ , and a wall thickness of the first segment **3121** is  $L_{sub.2}$ , satisfying a condition:  $L_{sub.2} > L_{sub.1}$ .

[0100] The condition,  $L_{\text{sub.2}} > L_{\text{sub.1}}$ , means that the minimum wall thickness of the first segment **3121** is greater than the wall thickness of the straight wall **311**, [0101] where the wall thickness  $L_{\text{sub.1}}$  of the straight wall **311** is the thickness size along the first direction X.

[0102] If the first segment **3121** is a plate-like structure extending along the first direction X, then the wall thickness  $L_{\text{sub.2}}$  of the first segment **3121** is the thickness size of the first segment **3121** along the second direction Y. If the first segment **3121** is a curved, arc-shaped structure, then the wall thickness  $L_{\text{sub.2}}$  of the first segment **3121** is the thickness size of the first segment **3121** in the direction of its radius of curvature, which means the wall thickness  $L_{\text{sub.2}}$  of the first segment **3121** is the thickness size of the first segment **3121** in a direction normal to a tangent line facing the surface of the holding space **313**.

[0103] By setting the wall thickness of the first segment **3121** of the bent wall **312** to be greater than the wall thickness of the straight wall **311**, the structural strength and the deformation resistance of the first segment **3121** of the bent wall **312** are increased. This effectively alleviates the occurrence of deformation or excessive curvature change of the first segment **3121** of the bent wall **312** when compressing the thermal management component **31** to absorb manufacturing tolerances, which is conducive to maintaining the original structural shape of the first segment **3121** of the bent wall **312**. Consequently, it can reduce the occurrence of excessive compression or crushing of the thermal management component **31**, thereby lowering the risk of the bent wall **312** being damaged or cracked during the compression process of the thermal management component **31**.

[0104] According to some embodiments of the present disclosure, with reference to FIG. 7 and FIG. 8, along the second direction Y, the first segment **3121** is an arc-shaped structure bending in a direction away from the holding space **313**.

[0105] That is to say, the first segment **3121** is an arc-shaped structure, and the first segment **3121** bends in the second direction Y away from the holding space **313**.

[0106] By arranging the first segment **3121** as an outwardly bending arc-shaped structure, on the one hand, it facilitates the connection between the first segment **3121** and the second segment **3122**, thus reducing the difficulty of manufacturing; on the other hand, it enables a smooth arc transition between the first segment **3121** and the second segment **3122**, which alleviates the phenomenon of excessively small curvature radius at the connection between the first segment **3121** and the second segment **3122**. This effectively reduces the risk of the connection between the first segment **3121** and the second segment **3122** being damaged or cracked during the compression process of the thermal management component **31**.

[0107] According to some embodiments of the present disclosure, with reference to FIG. 8, the wall thickness of the second segment **3122** gradually increases from the end connected to the straight wall **311** toward the end connected to the first segment **3121**.

[0108] A wall thickness of the end of the second segment **3122** connected to the straight wall **311** is the same as a wall thickness of the straight wall **311**.

[0109] By setting the wall thickness of the second segment **3122** to gradually increase from the end connected to the straight wall **311** to the end connected to the first segment **3121**, it facilitates manufacturing the first segment **3121** as a structure of which a wall thickness is greater than the wall thickness of the straight wall **311**, thus reducing manufacturing difficulty.

[0110] According to some embodiments of the present disclosure, referring to FIG. 9 and FIG. 10, FIG. 9 is a front view of a thermal management component **31** in a third direction Z provided in some embodiments of the present disclosure, and FIG. 10 is a partially enlarged view at C of a thermal management component **31** shown in FIG. 9. The thermal management component **31** further can include a reinforcement portion **314**, and the reinforcement portion **314** is arranged between the two second segments **3122** along the first direction X.

[0111] The reinforcement portion **314** is arranged between the two second segments **3122**, meaning that, along the first direction X, a reinforcement portion **314** is connected between the two second

segments **3122**, thus allowing the reinforcement portion **314** to provide support to the two second segments **3122**.

[0112] In one embodiment, there are multiple ways to connect the reinforcement portion **314**. For example, the reinforcement portion **314** can be connected to one of the two second segments **3122**, connected to both second segments **3122**, or connected to the first segment **3121**.

[0113] It should be noted that in the embodiments where the reinforcement portion **314** is provided between the two second segments **3122**, the wall thickness of the first segment **3121** can be set to be greater than the wall thickness of the straight wall **311**; or, the wall thickness of the first segment **3121** can be set to be equal to the wall thickness of the straight wall **311**. Exemplarily, in FIG. **10**, the wall thickness of the first segment **3121** is greater than the wall thickness of the straight wall **311**, and a reinforcement portion **314** is arranged between the two second segments **3122**.

[0114] By arranging a reinforcement portion **314** between the two second segments **3122** along the first direction X, the reinforcement portion **314** can provide certain support for the two second segments **3122**, thereby increasing the deformation resistance of the first segment **3121** of the bent wall **312**. This effectively alleviates the occurrence of deformation or excessive curvature change of the first segment **3121** of the bent wall **312** when compressing the thermal management component **31** to absorb manufacturing tolerances, which is conducive to maintaining the original structural shape of the first segment **3121** of the bent wall **312**. Consequently, it can reduce the occurrence of excessive compression or crushing of the thermal management component **31**, thereby lowering the risk of the bent wall **312** being damaged or cracked during the compression process of the thermal management component **31**.

[0115] In some embodiments, referring to FIGS. **9** and **10**, along the first direction X, one end of the reinforcement portion **314** is connected to one of the two second segments **3122**, and the other end is arranged at intervals with the other of the two second segments **3122**.

[0116] That is to say, the reinforcement portion **314** is connected only to one of the two second segments **3122** and has a gap with the other second segment **3122** along the first direction X.

[0117] By arranging the reinforcement portion **314** to be connected to one second segment **3122** and spaced from the other second segment **3122**, the reinforcement portion **314** can provide support to the two second segments **3122** and reserve compression allowance at the same time, thereby facilitating the compression of the thermal management component **31** to absorb manufacturing tolerances.

[0118] In some embodiments, referring to FIGS. **11** and **12**, FIG. **11** is a front view of a thermal management component **31** in a third direction Z provided in some embodiments of the present disclosure, and FIG. **12** is a partially enlarged view at D of a thermal management component **31** shown in FIG. **11**. Along the first direction X, each end of the reinforcement portion **314** is connected to one of the two second segments **3122**. In other words, the reinforcement portion **314** is connected to both second segments **3122**.

[0119] By connecting each end of the reinforcement portion **314** with each of the two second segments **3122**, this structure effectively enhances the connection strength between the reinforcement portion **314** and the bent wall **312**, which is beneficial for improving the supporting effect of the reinforcement portion **314** on the two second segments **3122**.

[0120] According to some embodiments of the present disclosure, with reference to FIG. **10** and FIG. **12**, along the second direction Y, the reinforcement portion **314** is arranged at intervals with the first segment **3121**. In other words, a gap is provided between the reinforcement portion **314** and the first segment **3121** in the second direction Y.

[0121] By arranging the reinforcement portion **314** at intervals with the first segment **3121** along the second direction Y, a gap is provided between the reinforcement portion **314** and the first segment **3121** in the second direction Y, which helps to reduce the manufacturing difficulty of the thermal management component **31**.

[0122] According to some embodiments of the present disclosure, with reference to FIGS. **6** and **7**,

and with further reference to FIG. 13, FIG. 13 is a partially enlarged view at E of a thermal management component **31** shown in FIG. 7. In some embodiments, the thermal management component **31** can further include multiple partition portions **315**. The multiple partition portions **315** are arranged at intervals along the second direction Y. The partition portion **315** is arranged obliquely between the two straight walls **311** and connected to the two straight walls **311**. The multiple partition portions **315** are configured to divide the holding space **313** into multiple medium flow channels **3131**, and the medium flow channels **3131** are configured for containing heat exchange medium.

[0123] The partition portion **315** is arranged obliquely between the two straight walls **311** and is connected to both straight walls **311**, meaning that the partition portion **315** forms an angle with the straight walls **311**. The two ends of the partition portion **315** in the first direction X are respectively connected to the two straight walls **311**. The partition portion **315** extends along the third direction Z, thus dividing the holding space **313** into multiple medium flow channels **3131** that extend along the third direction Z, and the multiple medium flow channels **3131** are arranged along the second direction Y.

[0124] By arranging multiple partition portions **315** at intervals along the second direction Y within the holding space **313** of the thermal management component **31**, the holding space **313** is divided into multiple medium flow channels **3131** configured for the flow of heat exchange medium. This facilitates the control of the flow rate and flow volume of the heat exchange medium within the thermal management component **31**. Additionally, by arranging the partition portion **315** in an inclined manner and connecting it between the two straight walls **311**, the partition portion **315** can deform more easily when the thermal management component **31** is compressed along the first direction X to absorb manufacturing tolerances, which helps to reduce the difficulty of compressing the thermal management component **31**.

[0125] According to some embodiments of the present disclosure, with reference to FIG. 13, the partition portion **315** is provided with an opposing first surface **3151** and second surface **3152** along its thickness direction. The two straight walls **311** include a first straight wall **3111** and a second straight wall **3112**, wherein the first straight wall **3111** is provided with a third surface **3111a** facing the medium flow channel **3131**, and the second straight wall **3112** is provided with a fourth surface **3112a** facing the medium flow channel **3131**. The first surface **3151** is arranged at an acute angle with the third surface **3111a** and at an obtuse angle with the fourth surface **3112a**, and the second surface **3152** is arranged at an obtuse angle with the third surface **3111a** and at an acute angle with the fourth surface **3112a**, wherein the first surface **3151** is connected to the third surface **3111a** through a first chamfered surface **316**, and/or the second surface **3152** is connected to the fourth surface **3112a** through a second chamfered surface **317**.

[0126] The first straight wall **3111** is provided with a third surface **3111a** facing the medium flow channel **3131**, and the second straight wall **3112** is provided with a fourth surface **3112a** facing the medium flow channel **3131**. In other words, the surface of the first straight wall **3111** facing the second straight wall **3112** in the first direction X is the third surface **3111a**, and the surface of the second straight wall **3112** facing the first straight wall **3111** in the first direction X is the fourth surface **3112a**.

[0127] The first surface **3151** is arranged at an acute angle with the third surface **3111a** and at an obtuse angle with the fourth surface **3112a**, meaning that the first surface **3151**, on one side of the partition portion **315** in its thickness direction, forms an acute angle with one straight wall **311** and an obtuse angle with the other. Similarly, the second surface **3152** is arranged at an obtuse angle with the third surface **3111a** and at an acute angle with the fourth surface **3112a**, meaning that the second surface **3152**, on the other side of the partition portion **315** in its thickness direction, forms an obtuse angle with one straight wall **311** and an acute angle with the other.

[0128] The first surface **3151** is connected to the third surface **3111a** via a first chamfered surface **316**, meaning that a chamfer is formed between the first surface **3151** of the partition portion **315**

on one side in its thickness direction and the third surface **3111a** of the first straight wall **3111** set at an acute angle. Similarly, the second surface **3152** is connected to the fourth surface **3112a** via a second chamfered surface **317**, meaning that a chamfer is formed between the second surface **3152** of the partition portion **315** on one side in its thickness direction and the fourth surface **3112a** of the second straight wall **3112** set at an acute angle.

[0129] Exemplarily, both the first chamfered surface **316** and the second chamfered surface **317** are curved surfaces. Of course, in other embodiments, the first chamfered surface **316** and the second chamfered surface **317** can also be flat surfaces.

[0130] The two surfaces of the partition portion **315** that form acute angles with the straight wall **311** are connected through chamfered surfaces, which increases the thickness of the portion where the partition portion **315** connects with the straight wall **311** along the second direction Y. On the one hand, it enhances the stability and reliability of the connection between the partition portion **315** and the straight wall **311**, and on the other hand, it also helps to alleviate the occurrence of cracks between the partition portion **315** and the straight wall **311**.

[0131] According to some embodiments of the present disclosure, with reference to FIG. 7 and FIG. 13, along the first direction X, a support portion **318** is provided in a protruding manner on a side of the straight wall **311** facing the holding space **313**, and a size of the support portion **318** is smaller than a distance between the two straight walls **311**.

[0132] The size of the support portion **318** is smaller than the distance between the two straight walls **311**, meaning that the support portion **318** protrudes from the surface of one straight wall **311** facing the other straight wall **311** and is arranged at intervals with the other straight wall **311** in the first direction X.

[0133] In one embodiment, one or multiple support portions **318** can be arranged on the straight wall **311**. Exemplarily, in FIG. 7, one support portion **318** is arranged on the straight wall **311**. If multiple support portions **318** are arranged on the straight wall **311**, the multiple support portions **318** are arranged at intervals along the second direction Y.

[0134] By providing a support portion **318** on the side of the straight wall **311** facing the holding space **313**, and with the size of the support portion **318** along the first direction X being smaller than the distance between the two straight walls **311**, the thermal management component **31** with this structure allows the support portion **318** to provide some support to the two straight walls **311** when the thermal management component **31** is compressed to absorb manufacturing tolerances, thereby alleviating the phenomenon of excessive compression or crushing of the thermal management component **31**. This is beneficial for maintaining the smooth flow of the heat exchange medium within the holding space **313**.

[0135] According to some embodiments of the present disclosure, with reference to FIG. 14, FIG. 14 is a front view of a thermal management component **31** in a third direction Z provided in some embodiments of the present disclosure. Both straight walls **311** are provided with support portions **318**, and along the first direction X, the projections of the support portions **318** on the two straight walls **311** do not overlap.

[0136] In the first direction X, the projections of the support portions **318** on the two straight walls **311** do not overlap, meaning that in the second direction Y, the support portions **318** arranged on the two straight walls **311** are positioned in a staggered manner.

[0137] It should be noted that in embodiments where the two straight walls **311** are provided with support portions **318**, one or multiple support portions **318** can be arranged on the straight wall **311**. Exemplarily, in FIG. 14, the two straight walls **311** are provided with multiple support portions **318**, and the multiple support portions **318** on each straight wall **311** are arranged at intervals along the second direction Y.

[0138] By providing support portions **318** on both straight walls **311**, where the projections of the support portions **318** on the two straight walls **311** along the first direction X do not overlap, the supporting effect of the support portions **318** on the two straight walls **311** is enhanced. This further



reduces the risk of excessive compression or crushing of the thermal management component **31**.  
[0139] According to some embodiments of the present disclosure, with reference to FIG. 3, FIG. 4, and FIG. 5, The embodiments of the present disclosure also provide a thermal management assembly **30**, wherein the thermal management assembly **30** includes a first current collector **32**, a second current collector **33**, and a thermal management component **31** as described above. Along the third direction Z, the first current collector **32** and the second current collector **33** are respectively connected to both ends of the thermal management component **31**. The first direction Y, the second direction Z, and the third direction Z are mutually perpendicular. The first current collector **32** and the second current collector **33** are each arranged with a medium inlet **321** and a medium outlet **331**, respectively, and both the medium inlet **321** and the medium outlet **331** communicate with the holding space **313**.

[0140] The first direction X is the thickness direction of the thermal management component **31**, the second direction Y is the width direction of the thermal management component **31**, and the third direction Z is the length direction of the thermal management component **31**.

[0141] In the embodiment where a partition portion **315** is arranged within the holding space **313** of the thermal management component **31**, the medium inlet **321** of the first current collector **32** and the medium outlet **331** of the second current collector **33** communicate with both ends of multiple medium flow channels **3131** in the third direction Z, thus enabling the function of collecting currents. This facilitates the simultaneous injection of the heat exchange medium into multiple medium flow channels **3131**, or it allows the heat exchange medium to flow out of multiple medium flow channels **3131** simultaneously.

[0142] The thermal management assembly **30** is further provided with a first current collector **32** and a second current collector **33**, which are connected to both ends of the thermal management component **31** along the third direction Z. The first current collector **32** and the second current collector **33** are respectively provided with a medium inlet **321** and a medium outlet **331**. Therefore, by communicating the two ends of the holding space **313** respectively with the medium inlet **321** and the medium outlet **331**, the heat exchange medium can flow into or out of the holding space **313**, thereby enabling the thermal management function of the thermal management assembly **30**.

[0143] According to some embodiments of the present disclosure, with reference to FIG. 2 and FIG. 3, The present disclosure also provides a battery **100**, wherein the battery **100** includes a box **10**, a battery cell **20**, and any of the thermal management assemblies **30** in the above solutions. The battery cell **20** is accommodated within the box **10**, and the thermal management assembly **30** is arranged within the box **10**, wherein the thermal management assembly **30** is configured to regulate the temperature of the battery cell **20**.

[0144] Exemplarily, in FIG. 3, the battery cell **20** has a rectangular shape. Multiple battery cells **20** are arranged in a layout, wherein the battery **100** includes multiple rows of battery cells **20**, and the multiple rows of battery cells **20** are arranged along a first direction X, with each row of battery cells **20** including multiple battery cells **20** arranged along a third direction Z.

[0145] According to some embodiments of the present disclosure, with reference to FIG. 3, the battery cell **20** is provided with a first side surface **21**, wherein the first side surface **21** is a surface with the largest area among the outer surfaces of the battery cell **20**, and the first side surface **21** abuts the straight wall **311**.

[0146] The first side surface **21** abuts the straight wall **311** of the thermal management component **31**, which can be a direct abutment between the first side surface **21** and the straight wall **311** of the thermal management component **31**, meaning that the face with the largest area of the outer surfaces of the battery cell **20** is directly abutted to the straight wall **311** of the thermal management component **31**. Of course, in some embodiments, it can also be that the first side surface **21** indirectly abuts the straight wall **311** of the thermal management component **31**, meaning that the face with the largest area of the outer surfaces of the battery cell **20** is abutted to the straight wall

**311** of the thermal management component **31** through a thermal conductive member, such as thermal adhesive or thermal pad.

[0147] It should be noted that the battery cell **20** is of a rectangular structure, and the first side surface **21** is the surface of the battery cell **20** with the largest area among the outer surfaces, meaning that the first side surface **21** is the outer surface of the battery cell **20** in the thickness direction of the battery cell **20**. In other words, the thermal management assembly **30** is arranged between two adjacent rows of battery cells **20** along the thickness direction of the battery cell **20**, and the straight wall **311** of the thermal management component **31** of the thermal management assembly **30** abuts the first side surface **21** of the battery cell **20** to achieve thermal exchange between the battery cell **20** and the thermal management component **31**.

[0148] The first direction X is consistent with the thickness direction of the battery cell **20**, the second direction Y is consistent with the width direction of the battery cell **20**, and the third direction Z is consistent with the length direction of the battery cell **20**.

[0149] By arranging the straight wall **311** of the thermal management component **31** to abut the first side surface **21** of the battery cell **20**, that is, by arranging the thermal management component **31** on the side of the battery cell **20** with the largest surface area, sufficient heat exchange area between the battery cell **20** and the thermal management component **31** can be ensured. This is beneficial for enhancing the thermal management capability of the thermal management component **31** for the battery cell **20**, thereby effectively reducing the safety risks caused by temperature rise during the use of the battery **100**.

[0150] According to some embodiments of the present disclosure, the embodiments of the present disclosure provide an electric device, wherein the electric device includes a battery **100** of any of the above solutions, and the battery **100** is configured to provide electrical energy to the electric device.

[0151] The electric device can be any of the above-mentioned devices or systems that apply the battery **100**.

[0152] According to some embodiments of the present disclosure, with reference to FIGS. 4 to 8, and FIG. 13, The present disclosure provides a thermal management component **31**, wherein the thermal management component **31** includes two straight walls **311**, two bent walls **312**, and multiple partition portions **315** and support portions **318**. The two straight walls **311** are arranged oppositely along a first direction X. Along the second direction Y, each of the two bent walls **312** is connected to one end of the straight walls **311**, and the bent wall **312** bends in a direction away from the holding space **313**, wherein the second direction Y is perpendicular to the first direction X. The two bent walls **312** and the two straight walls **311** enclose to form a holding space **313**, and the holding space **313** is configured for containing a heat exchange medium. The bent wall **312** includes a first segment **3121** and two second segments **3122**. A second segment **3122**, a first segment **3121**, and the other second segment **3122** are sequentially connected. The first segment **3121** is an arc-shaped structure that bends in the second direction Y away from the holding space **313**. The two second segments **3122** are respectively connected to the two straight walls **311**. The second segment **3122** is arranged obliquely relative to the straight wall **311** connected to it, and along the first direction X, the first segment **3121** is positioned between the two straight walls **311**. The wall thickness of the straight wall **311** is greater than the wall thickness of the first segment **3121**, and the wall thickness of the second segment **3122** gradually increases from the end connected to the straight wall **311** toward the end connected to the first segment **3121**. The multiple partition portions **315** are arranged at intervals along the second direction Y. The partition portion **315** is arranged obliquely between the two straight walls **311** and connected to the two straight walls **311**. The multiple partition portions **315** are configured to divide the holding space **313** into multiple medium flow channels **3131**, and the medium flow channels **3131** are configured for containing heat exchange medium. The two surfaces of the partition portion **315** that form acute angles with the straight wall **311** are connected through chamfered surfaces. Along the first

direction X, a support portion **318** is provided in a protruding manner on a side of the straight wall **311** facing the holding space **313**, and a size of the support portion **318** is smaller than a distance between the two straight walls **311**.

[0153] It should be noted that, without conflicting, the embodiments and features described in the embodiments of the present disclosure can be combined with each other.

[0154] The above are only preferred embodiments of the present disclosure, which is not intended to limit the present disclosure, and the present disclosure can have various changes and variations for those skilled in the art. Any modification, equivalent substitution, improvement, etc. made within the spirit and principles of the present disclosure shall be included in the scope of protection of the present disclosure.

## Claims

1. A thermal management component, comprising two straight walls, arranged oppositely along a first direction; and two bent walls, each connected to one end of the straight walls along a second direction, wherein the second direction is perpendicular to the first direction; and the two bent walls and the two straight walls enclose to form a holding space configured to contain a heat exchange medium.
2. The thermal management component according to claim 1, wherein, along the second direction, the bent walls bend in a direction away from the holding space.
3. The thermal management component according to claim 2, wherein each of the two bent walls comprises: a first segment; a second segment; and an other second segment, wherein the first segment, the second segment, and the other second segment are sequentially connected; the two second segments are respectively connected to the two straight walls; the second segment is arranged obliquely relative to the straight wall connected to it; and the first segment is positioned between the two straight walls along the first direction.
4. The thermal management component according to claim 3, wherein a wall thickness of the straight wall is  $L_{sub.1}$ , and a wall thickness of the first segment is  $L_{sub.2}$ , satisfying a condition:  $L_{sub.2} > L_{sub.1}$ .
5. The thermal management component according to claim 4, wherein along the second direction, the first segment is an arc-shaped structure bending in a direction away from the holding space.
6. The thermal management component according to claim 4, wherein a wall thickness of the second segment gradually increases from an end connected to the straight wall toward an end connected to the first segment.
7. The thermal management component according to claim 3, wherein the thermal management component further comprises: a reinforcement portion that is arranged between the two second segments along the first direction.
8. The thermal management component according to claim 7, wherein an end of the reinforcement portion is connected to one of the two second segments along the first direction; and another end of the reinforcement portion is arranged at intervals with the second segment and the other second segment.
9. The thermal management component according to claim 7, wherein, along the first direction, each end of the reinforcement portion is connected to at least one of the second segment or the other second segment.
10. The thermal management component according to claim 7, wherein, along the second direction, the reinforcement portion is arranged at intervals with the first segment.
11. The thermal management component according to claim 1, wherein the thermal management component further comprises: a plurality of partition portions arranged at intervals along the second direction; the plurality of partition portions are arranged obliquely between the two straight walls and connected to the two straight walls; the plurality of partition portions divide the holding

space into a plurality of medium flow channels; and the plurality of medium flow channels are configured to contain heat exchange mediums.

**12.** The thermal management component according to claim 11, wherein a partition portion comprises an opposing first surface and second surface along its thickness direction; the two straight walls comprise a first straight wall and a second straight wall, wherein the first straight wall comprises a third surface facing a medium flow channel; and the second straight wall comprises a fourth surface facing the medium flow channel; the first surface is arranged at an acute angle with the third surface and at an obtuse angle with the fourth surface; the second surface is arranged at an obtuse angle with the third surface and at an acute angle with the fourth surface; and the first surface is connected to the third surface through a first chamfered surface; or the second surface is connected to the fourth surface through a second chamfered surface.

**13.** The thermal management component according to claim 1, wherein a support portion protrudes on a side of the straight wall facing the holding space along the first direction; and a size of the support portion is smaller than a distance between the two straight walls.

**14.** The thermal management component according to claim 13, wherein the two straight walls comprise support portions; and projections of the support portions on the two straight walls do not overlap along the first direction.

**15.** A thermal management assembly, comprising a thermal management component, the thermal management component comprising: two straight walls, arranged oppositely along a first direction; two bent walls, each connected to one end of the straight walls along a second direction, wherein the second direction is perpendicular to the first direction; and the two bent walls and the two straight walls enclose to form a holding space configured to contain a heat exchange medium; a first current collector comprising a medium inlet that is in communication with the holding space; and a second current collector comprising a medium outlet that is in communication with the holding space, wherein the first current collector and the second current collector are respectively connected to two ends of the thermal management component along a third direction; and the first direction, the second direction, and the third direction are perpendicular to each other.

**16.** A battery, comprising a box; a battery cell housed within the box; and a thermal management assembly arranged in the box and configured to manage a temperature of the battery cell, the thermal management assembly comprising a thermal management component, the thermal management component comprising: two straight walls, arranged oppositely along a first direction; two bent walls, each connected to one end of the straight walls along a second direction, wherein the second direction is perpendicular to the first direction; and the two bent walls and the two straight walls enclose to form a holding space configured to contain a heat exchange medium; a first current collector comprising a medium inlet that is in communication with the holding space; and a second current collector comprising a medium outlet that is in communication with the holding space, wherein the first current collector and the second current collector are respectively connected to two ends of the thermal management component along a third direction; and the first direction, the second direction, and the third direction are perpendicular to each other.

**17.** The battery according to claim 16, wherein the bent walls bend in a direction away from the holding space along the second direction.

**18.** The battery according to claim 16, wherein each of the two bent walls comprises: a first segment; a second segment; and another second segment, wherein the first segment, the second segment, and the other second segment are sequentially connected; the two second segments are respectively connected to the two straight walls; the second segment is arranged obliquely relative to the straight wall connected to it; and the first segment is positioned between the two straight walls along the first direction.

**19.** The battery according to claim 16, wherein the battery cell comprises a first side surface with a largest area among outer surfaces of the battery cell and abuts the straight wall.

**20.** An electric device, comprising the battery according to claim 16, wherein the battery is configured to provide electrical energy.

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